

Lecture 3 Real Business Cycle Model

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Basic RBC model: Social Planner's Problem.

Preference: $U = E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\bar{\sigma}} - 1}{1-\bar{\sigma}} \right]$

expectation given information at $t=0$ $\star$ Produced goods & depreciated capital in last

Technology: $C_t + k_t = Z_t \bar{K}_{t-1}^\alpha N_t^{1-\alpha} + (1-\delta)K_{t-1}$ period used for consumption & new capital
the amount of capital is predetermined in last period. In current period t ,
 Z_t is the TFP which follows the process: agent decide on K_t .

$$\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t$$

steady state TFP, usually normalize $\bar{Z}=1 \Rightarrow \log \bar{Z}=0$

Endowment:

$$N_t = 1, \quad K_{-1} > 0$$

time/labor endowment initial capital stock

Information: decision made based on all information I_t up to t

Social planner maximize the social utility by choosing C_t, K_t

$$\max_{C_t, K_t} U \equiv \max_{C_t, K_t} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\bar{\sigma}} - 1}{1-\bar{\sigma}} \right]$$

Subject to a budget constraint:

$$\forall t, \quad C_t + k_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1-\delta)K_{t-1}$$

$$N_t = 1, \quad K_{-1} > 0$$

Set up dynamic Lagrangian equation:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\bar{\sigma}} - 1}{1-\bar{\sigma}} + \lambda_t [Z_t K_{t-1}^\alpha + (1-\delta)K_{t-1} - C_t - k_t] \right]$$

F.O.C. $\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow C_t^{-\bar{\sigma}} - \lambda_t = 0. \quad (1)$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \Rightarrow \lambda_t \beta + E_t [\lambda_{t+1} \beta^{t+1} (\alpha Z_{t+1} K_t^{\alpha-1} + 1-\delta)] = 0.$$

$$\Rightarrow -\lambda_t + \beta E_t [\lambda_{t+1} (\alpha Z_{t+1} K_t^{\alpha-1} + 1-\delta)] = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow C_t + k_t - Z_t K_{t-1}^\alpha - (1-\delta)K_{t-1} = 0$$

Transversality Condition: $\lim_{t \rightarrow \infty} \beta^t E_t \lambda_t K_t = 0$

Plug in (1) into (2) $\Rightarrow \beta E_t [C_{t+1}^{-\bar{\sigma}} (\alpha Z_{t+1} K_t^{\alpha-1} + 1-\delta)] = C_t^{-\bar{\sigma}}$

$$\Rightarrow 1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\bar{\sigma}} (\alpha Z_{t+1} K_t^{\alpha-1} + 1-\delta) \right]$$

We have a dynamic system of (K_{t+1}, C_t, Z_{t+1})

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$$K_t = Z_t K_{t-1}^\alpha + (1-\delta)K_{t-1} - C_t$$

$$1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{1-\sigma}} (1-\delta + \alpha Z_{t+1} K_t^{\alpha-1}) \right]$$

two state variables

$$\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{iid } N(0, \sigma^2)$$

one shock variable

Three variables, three equations: $\#(\text{equation}) = \#(\text{variable})$

* What is the economic intuition behind Lagrangian multiplier?

Lagrangian multiplier is the shadow price of constrained resources. Marginal value of relaxing or tightening a constraint. For a problem,

$$V(b) = \max_x f(x) \quad \text{subject to } g(x) \leq b$$

$$\mathcal{L}(x, \lambda) = f(x) + \lambda (b - g(x))$$

* If constraint is binding $g(x) = b$, λ shows how much the objective function would increase if you relax the constraint slightly

Envelope theorem: $\frac{\partial V(b)}{\partial b} = \frac{\partial \mathcal{L}(x, \lambda; b)}{\partial b} \Big|_{x^*, \lambda^*} \Rightarrow V'(b) = \lambda^*(b)$

$$\frac{\partial \mathcal{L}}{\partial b} = \lambda$$

$$V(b+\Delta) - V(b) \approx \lambda^*(b) \Delta$$

* If constraint is non-binding $g(x) < b$, $\lambda^* = 0 \Rightarrow V'(b) = 0$

Shadow value

A relaxation of constraint will not improve the objective function.

eg: RBC: $\max_{\{C_{t+j}, K_{t+j+1}\}} E_t \sum_{j=0}^{\infty} \beta^j \frac{C_{t+j}^{1-\sigma}}{1-\sigma}$ subject to $C_{t+j} + K_{t+j+1} = F(Z_{t+j}, K_{t+j}) + (1-\delta)K_{t+j} + a_t$

$$\mathcal{L} = E_t \sum_j \beta^j \left(\frac{C_{t+j}^{1-\sigma}}{1-\sigma} + \lambda_{t+j} [F(\cdot) + (1-\delta)K_{t+j} - C_{t+j} - K_{t+j+1} + a_{t+j}] \right)$$

+ a_t

appear only at $j=0$,

perturb only one period's constraint

$$\frac{\partial W_t}{\partial a_t} = \frac{\partial \mathcal{L}_t}{\partial a_t} \Big|_{\text{at optimum}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t C_t^{-\sigma} - \beta^t \lambda_t = 0 \Rightarrow C_t^{-\sigma} = \lambda_t$$

$$\Rightarrow \frac{\partial W_t}{\partial a_t} = \lambda_t = C_t^{-\sigma} = U_C(C_t) = \frac{\partial (\text{max attainable lifetime utility from } t)}{\partial (\text{one more unit of resources at } t)}$$

* What is transversality condition (TVC)?

Boundary condition at infinity that rules out Ponzi-scheme: leaving valuable resources unspent forever

In a typical RBC model. TVC: $\lim_{T \rightarrow \infty} \beta^T U_C(C_T) K_{T+1} = 0$ (15)

The discounted marginal utility value of the capital stock carried into the far future must vanish. Otherwise you're wasting utility by not consuming some of it earlier.

At large T , if we increase C_T by ϵ and decrease future capital by ϵ .

$$\Rightarrow \left. \begin{array}{l} \text{First-order loss in capital: } \beta^{T+1} \lambda_{T+1} \epsilon \\ \text{First-order gain in utility: } \beta^T U_C \epsilon \end{array} \right\} \Rightarrow \lim_{T \rightarrow \infty} \beta^T U_C K_{T+1} = 0$$

→ in far future, you can not gain by this operation

9 Basic RBC model: competitive market approach.

* divide social planner into household & firm.

Household's problem: Household's own capital K_{t+1} & one unit of labor $N_t = 1$. They lend capital and labor to firms and earn interest R_t & wage W_t .

$$\max_{\{C_t, K_t, N_t\}_{t=0}^{\infty}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\bar{\sigma}} - 1}{1-\bar{\sigma}} \right]$$

$$\text{s.t. } C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta) K_{t-1}^{(s)} - K_{t-1}^{(s)} \text{ given, budget constraint}$$

→ (s) means supply

$$\text{No-ponzi Scheme condition: } \lim_{t \rightarrow \infty} E_0 \left[\left(\prod_{s=0}^t R_s \right) K_t^{(s)} \right] = 0.$$

(TVC)

Firm's problem: Representative firm with tech Z_t , hire capital & labor to produce, and sells output to maximize profit. Given W_t, R_t , firm chooses $N_t^{(d)}$ & $K_t^{(d)}$

$$\text{profit } \pi_t = \max_{\{K_t^{(d)}, N_t^{(d)}\}} Z_t K_t^{(d)\alpha} N_t^{(d)1-\alpha} - W_t N_t^{(d)} - R_t K_t^{(d)}$$

→ (d) means demand

where tech follows AR(1): $\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \epsilon_t, \epsilon_t \sim \text{iid } N(0, \sigma^2)$

Solve household's problem with Lagrangian Multiplier Method:

$$\mathcal{L} = E_0 \left[\sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\bar{\sigma}} - 1}{1-\bar{\sigma}} + \lambda_t (C_t + K_t^{(s)} - W_t N_t^{(s)} - R_t K_{t-1}^{(s)} - (1-\delta) K_{t-1}^{(s)}) \right] \right]$$

$$\Rightarrow \text{F.O.C. } \frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow C_t^{-\bar{\sigma}} - \lambda_t = 0. \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \Rightarrow -\beta^t \lambda_t + \beta^{t+1} E_t [\lambda_{t+1} (R_{t+1} + 1 - \delta)] = 0 \Rightarrow \lambda_t = \beta E_t [\lambda_{t+1} (R_{t+1} + 1 - \delta)]$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta) K_{t-1}^{(s)} \quad (2)$$

Euler equations. Plug (1) into (2) $\Rightarrow C_t^{-\bar{\sigma}} = \beta E_t [C_{t+1}^{-\bar{\sigma}} (R_{t+1} + 1 - \delta)]$

return of extra unit of K tomorrow

Marginal utility of 1 unit C today $\leftarrow$ discount to today $\leftarrow$ Marginal utility of 1 unit C tomorrow.

LHS: The utility of 1 unit of consumption today.

RHS: The expected utility of sacrificing 1 unit of consumption today, instead save as capital and carry over to next period, get R_{t+1} (rent earning) & get $(1-\delta)$ unit of depreciated capital, consume $R_{t+1} + 1-\delta$ in the next period at marginal utility of consumption $\lambda_{t+1} = C_{t+1}^{-\alpha}$, Finally, discount to current value by multiplying β .

LHS = RHS $\Rightarrow$ Intertemporal tradeoff (no "arbitrage")

* A TFP shock on z_t change the level through resource constraint, and dynamics through the Euler equation. Resource constraint: $C_t + k_t = z_t k_{t-1}^\alpha N_t^{1-\alpha} + (1-\delta)k_{t-1}$.

Solve firm's problem with Lagrangian multiplier method:

$$\frac{\partial (z_t (k_{t-1}^{(d)})^\alpha (N_t^{(d)})^{1-\alpha} - W_t N_t^{(d)} - R_t k_{t-1}^{(d)})}{\partial k_{t-1}^{(d)}} = 0.$$

$$\Rightarrow \alpha z_t (k_{t-1}^{(d)})^{\alpha-1} (N_t^{(d)})^{1-\alpha} = R_t \quad (4) \quad MPK = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t^{(d)}} = 0 \Rightarrow (1-\alpha) z_t (k_{t-1}^{(d)})^\alpha (N_t^{(d)})^{-\alpha} = W_t \quad (5) \quad MPL = W_t$$

Close the model: Market clearing conditions

Capital market: $k_{t-1}^{(d)} = k_{t-1}^{(s)}$... capital demand = supply

Labor market: $N_t^{(d)} = N_t^{(s)} = 1$... labor demand = supply

Goods market: $C_t + k_t = z_t k_{t-1}^\alpha N_t^{1-\alpha} + (1-\delta)k_{t-1}$... Resource constraint.

Walras Law: If there are m markets, then we need $m-1$ number of market clearing conditions, The last market is automatically cleared.

Thus, here we need two out of three. We keep resource constraint & capital market clearing by dropping (d) & (s). We also plug in constant labor supply & demand.

How resource constraint is derived?

$$\text{Competitive market} \Rightarrow \Pi_t = 0 \Rightarrow z_t (k_{t-1}^{(d)})^\alpha (N_t^{(d)})^{1-\alpha} = W_t N_t^{(d)} + R_t k_{t-1}^{(d)}$$

$$(3) \text{ from household's problem} \Rightarrow C_t + k_t^{(s)} = W_t N_t^{(s)} + R_t k_{t-1}^{(s)} + (1-\delta)k_{t-1}$$

$$(s) = (d) \Rightarrow C_t + k_t = z_t k_{t-1}^\alpha N_t^{1-\alpha} + (1-\delta)k_{t-1} \quad * \text{Produced goods + depreciated capital from last period is divided into use of capital and consumption}$$

System of equations of the competitive market version of the model:

$$(i) k_t = z_t k_{t-1}^\alpha + (1-\delta)k_{t-1} - C_t \quad \dots \text{resource constraint}$$

$$(ii). 1 = E_t \left[\beta \left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{\sigma}} \left[\alpha Z_{t+1} K_t^{\alpha-1} + (1-\delta) \right] \right]$$

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$$(iii). \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim iid N(0, \sigma^2)$$

* The competitive market version is equivalent to social planner's version. (isomorphic)

⇒ Same equilibrium outcome.

* The competitive market version of RBC model is a decentralized version of social planner RBC.

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6 First and second welfare theorem in dynamics equilibrium:

First welfare theorem: Provided that all product and factor markets are competitive and there are no externalities in production or consumption, dynamic competitive equilibria are Pareto Optimal. *no "friction" world in economics*

Second welfare theorem: All Pareto optimal allocation can be decentralized as a dynamic competitive equilibrium.

- 1st Welfare theorem in RBC: CE allocation solves the social planner's problem ⇒ Competitive equilibria are Pareto Optimal
- 2nd welfare theorem in RBC: Social planner allocation solves the competitive equilibrium ⇒ Pareto efficient allocation is supported by CE. (∃ competitive prices that support)

Both theorems rely on:

- ① perfect competition (price taking)
- ② No externalities, no taxes/distortions, no market power.
- ③ Complete contracting, no missing markets (rep agent RBC avoids this)
- ④ Concave utility, convex technology set (CRS + concavity)
- ⑤ TVC / no-Ponzi to rule out pathological asset plans

Question 1. If there is a exogenous TFP shock Z_t , which moves more on impact C_t or K_t ?

* Typically K_t moves more (investment is more volatile than consumption)

$$\Delta C_t + \Delta K_t = \Delta Y_t$$

$$\text{In CRRA case: } U_C(C_t) = \beta E_t [U_C(C_{t+1}) R_{t+1}] \Rightarrow 1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{\sigma}} R_{t+1} \right]$$

$Z_t \uparrow \Rightarrow MPK \uparrow \Rightarrow R_{t+1} \uparrow \Rightarrow$ By Euler equation, consumption growth: $\frac{C_{t+1}}{C_t} = (\beta R_{t+1})^{\frac{1}{\sigma}}$
 $\Rightarrow R_{t+1} \uparrow \Rightarrow \frac{C_{t+1}}{C_t} \uparrow \Rightarrow$ consumption growth higher, future consumption is higher relative to today.

$\Delta Y_t = \Delta C_t + \Delta K_t$, if all extra resources ΔY_t go to ΔC_t , then not enough investment, next period consumption can not boost much ⇒ keep C_t from not jumping too much, increase investment to make C_{t+1} higher.

* Persistent TFP shock will make this story even stronger.

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Question 2: What if households sell capital to firms and firms sell back to household after production?

Household's problem:

$$\max_{\{C_t, k_t, N_t\}_{t=0}^{\infty}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\bar{\sigma}}}{1-\bar{\sigma}} \right]$$

$$\text{s.t. } C_t + k_t^{(s)} = W_t N_t^{(s)} + R_t k_{t-1}^{(s)}$$

$$\Rightarrow \begin{cases} C_t^{-\bar{\sigma}} = \beta E_t [C_{t+1}^{-\bar{\sigma}} R_{t+1}] & (1) \\ C_t + k_t^{(s)} = W_t N_t^{(s)} + R_t k_{t-1}^{(s)} & (2) \end{cases}$$

Firm's problem:

$$\max_{\{k_{t-1}^{(d)}, N_t^{(d)}\}} \Pi_t = Z_t (k_{t-1}^{(d)})^\alpha (N_t^{(d)})^{1-\alpha} - W_t N_t^{(d)} - R_t k_{t-1}^{(d)} + (1-\delta) k_{t-1}^{(d)}$$

$$\Rightarrow \begin{cases} \alpha Z_t (k_{t-1}^{(d)})^{\alpha-1} (N_t^{(d)})^{1-\alpha} + (1-\delta) = R_t & (3) \\ (1-\alpha) Z_t (k_{t-1}^{(d)})^\alpha (N_t^{(d)})^{-\alpha} = W_t & (4) \end{cases}$$

System of equations is exactly the same as SP's problem. (isomorphic)

Here, R_t is no longer rental rate, but the gross return factor: rental + resale value.

Define the usual rental rate as $r_t = R_t - (1-\delta)$

$$\begin{cases} (3) \Rightarrow MPK = r_t \\ \Pi_t = Z_t K^\alpha N^{1-\alpha} + (1-\delta)K - RK - WN = Z_t K^\alpha N^{1-\alpha} - (R - (1-\delta))K - WN \\ \quad = Z_t K^\alpha N^{1-\alpha} - rK - WN \end{cases}$$

$$\begin{cases} (1) \Rightarrow C_t^{-\bar{\sigma}} = \beta E_t [C_{t+1}^{-\bar{\sigma}} (R_{t+1} + (1-\delta))] \\ (2) \Rightarrow C_t + k_t = W_t N_t + R_t k_{t-1} + (1-\delta) k_{t-1} \end{cases}$$

$\Rightarrow$ Setting is totally isomorphic.

6 Solve the RBC model.

Find the Steady State (S.S.) of the economy, drop time subscript.

$$\bar{C} = \bar{Z} \bar{K}^\alpha + (1-\delta)\bar{K} - \bar{K} \quad \dots \text{Resource constraint.} \quad C_t = C_{t+1} = \bar{C}$$

$$1 = \beta \cdot [\alpha \bar{Z} \bar{K}^{\alpha-1} + (1-\delta)] \quad \dots \text{Euler equation}$$

$$K_t = K_{t+1} = \bar{K}$$

$$Z_t = Z_{t+1} = \bar{Z}$$

$$N_t = 1, \forall t$$

Two variables. $(\bar{K}, \bar{C})$ given parameters $(\bar{Z}, \beta, \alpha, \delta)$

$$\bar{K} = \left[\frac{\alpha \bar{Z}}{\frac{1}{\beta} - (1-\delta)} \right]^{\frac{1}{1-\alpha}}, \quad \bar{C} = \bar{Z} \bar{K}^\alpha - \delta \bar{K}, \quad \bar{Y} = \bar{Z} \bar{K}^\alpha$$

$$\frac{\bar{C}}{\bar{K}} = \bar{Z} \bar{K}^{\alpha-1} - \delta = \frac{\bar{Y}}{\bar{K}} - \delta = \frac{\frac{1}{\beta} - (1-\delta)}{\alpha} - \delta = \frac{\frac{1}{\beta} - 1 + (1-\alpha)\delta}{\alpha}$$

Log-linearization:

(i) Resource Constraint: $K_t = Z_t K_{t-1}^\alpha + (1-\delta)K_{t-1} - C_t \Rightarrow \bar{K} = \bar{Z} \bar{K}^\alpha + (1-\delta)\bar{K} - \bar{C}$

$$\bar{K}(1 + \hat{k}_t) = \bar{Z}(1 + \hat{z}_t) \bar{K}^\alpha (1 + \hat{k}_{t-1})^\alpha + (1-\delta)\bar{K}(1 + \hat{k}_{t-1}) - \bar{C}(1 + \hat{c}_t)$$

$$\Rightarrow \bar{K} + \bar{K} \hat{k}_t = \bar{Z} \bar{K}^\alpha (1 + \hat{z}_t)(1 + \alpha \hat{k}_{t-1}) + (1-\delta)\bar{K} + (1-\delta)\bar{K} \hat{k}_{t-1} - \bar{C} - \bar{C} \hat{c}_t$$

$$\Rightarrow \bar{K} + \bar{K} \hat{k}_t = \bar{Z} \bar{K}^\alpha + \alpha \bar{Z} \bar{K}^\alpha \hat{k}_{t-1} + \bar{Z} \bar{K}^\alpha \hat{z}_t + \bar{Z} \bar{K}^\alpha \alpha \hat{k}_{t-1} \hat{z}_t + (1-\delta)\bar{K} + (1-\delta)\bar{K} \hat{k}_{t-1} - \bar{C} - \bar{C} \hat{c}_t$$

$$\Rightarrow \bar{K} \hat{k}_t = \frac{\alpha \bar{Z} \bar{K}^\alpha}{\bar{Y}} \hat{k}_{t-1} + \frac{\bar{Z} \bar{K}^\alpha}{\bar{Y}} \hat{z}_t + (1-\delta)\bar{K} \hat{k}_{t-1} - \bar{C} \hat{c}_t$$

$$\Rightarrow \hat{k}_t = \frac{\alpha \bar{Y}}{\bar{K}} \hat{k}_{t-1} + \frac{\bar{Y}}{\bar{K}} \hat{z}_t + (1-\delta) \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t$$

$$\Rightarrow \hat{k}_t = \underbrace{\left(\alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta \right)}_{\textcircled{1}} \hat{k}_{t-1} + \underbrace{\frac{\bar{Y}}{\bar{K}} \hat{z}_t}_{\textcircled{2}} - \underbrace{\frac{\bar{C}}{\bar{K}} \hat{c}_t}_{\textcircled{3}}$$

* next period change in k equals ① change in k_{t-1} that result in change in output $\left(\alpha \frac{\bar{Y}}{\bar{K}} \right)$ & depreciated capital $1-\delta$
 ② change in tech z that result in change in output
 ③ Change in consumption C that result in change in investment.

(ii) Euler equation: $1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{\sigma}} (1-\delta + \alpha Z_{t+1} K_t^{\alpha-1}) \right]$

Define $R_{t+1} = 1 - \delta + \alpha Z_{t+1} K_t^{\alpha-1} \Rightarrow \bar{R} = \frac{1}{\beta}$

from $1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{\sigma}} R_{t+1} \right] \Rightarrow \frac{1}{\beta} = \bar{R} = E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{\sigma}} R_{t+1} \right]$

$$\Rightarrow 1 = E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\frac{1}{\sigma}} \frac{R_{t+1}}{\bar{R}} \right] \Rightarrow 1 = E_t \left[\left(\frac{\bar{C} e^{\hat{c}_t}}{\bar{C} e^{\hat{c}_{t+1}}} \right)^{\frac{1}{\sigma}} e^{\hat{r}_{t+1}} \right]$$

$$\Rightarrow 1 = E_t \left[e^{\frac{1}{\sigma} (\hat{c}_t - \hat{c}_{t+1})} + \hat{r}_{t+1} \right] \quad \text{Use } e^x \approx 1+x \text{ for small } x$$

$$\Rightarrow 1 \approx E_t \left[1 + \frac{1}{\sigma} (\hat{c}_t - \hat{c}_{t+1}) + \hat{r}_{t+1} \right] \Rightarrow \frac{1}{\sigma} \hat{c}_t = \frac{1}{\sigma} E_t \hat{c}_{t+1} - E_t \hat{r}_{t+1} \quad \textcircled{1}$$

Now derive $\hat{r}_{t+1}$. Let $MPK_{t+1} = \alpha Z_{t+1} K_t^{\alpha-1} \Rightarrow \overline{MPK} = \alpha \bar{Z} \bar{K}^{\alpha-1}$

$$\Rightarrow \bar{R} = 1 - \delta + \overline{MPK} \quad \text{Now approximate } MPK_{t+1} \text{ around steady state.}$$

$$\alpha Z_{t+1} K_t^{\alpha-1} \approx \alpha \bar{Z} (1 + \hat{z}_{t+1}) \bar{K}^{\alpha-1} (1 + \hat{k}_t)^{\alpha-1} = \alpha \bar{Z} \bar{K}^{\alpha-1} (1 + \hat{z}_{t+1}) [1 + (\alpha-1) \hat{k}_t]$$

$$= \overline{MPK} [1 + \hat{z}_{t+1} + (\alpha-1) \hat{k}_t]$$

$$R_{t+1} \approx (1-\delta) + \overline{MPK} + \overline{MPK} (\hat{z}_{t+1} + (\alpha-1)\hat{k}_t) = \overline{R} + \overline{MPK} (\hat{z}_{t+1} + (\alpha-1)\hat{k}_t) \quad (20)$$

$$\hat{r}_{t+1} = \log\left(\frac{R_{t+1}}{\overline{R}}\right) \approx \frac{R_{t+1} - \overline{R}}{\overline{R}} = \frac{\overline{MPK}}{\overline{R}} (\hat{z}_{t+1} + (\alpha-1)\hat{k}_t)$$

$$\underbrace{\frac{\overline{MPK}}{\overline{R}}}_{\text{gross return}} = \frac{\overline{R} - (1-\delta)}{\overline{R}} = 1 - \frac{1-\delta}{\beta} = 1 - (1-\delta)\beta$$

$$\Rightarrow \hat{r}_{t+1} = [1 - (1-\delta)\beta] [\hat{z}_{t+1} - (1-\alpha)\hat{k}_t] \quad (21)$$

Plug (21) into (1) $\Rightarrow \quad \delta \hat{c}_t = \delta E_t \hat{c}_{t+1} + (1-\alpha)[1 - (1-\delta)\beta] \hat{k}_t - [1 - (1-\delta)\beta] E_t \hat{z}_{t+1}$

Let $\tilde{\delta} = 1 - (1-\delta)\beta$ share of gross return that comes from new output rather than keeping old capital.

$$\Rightarrow \quad \delta \hat{c}_t = \delta E_t \hat{c}_{t+1} + \tilde{\delta} (1-\alpha) \hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1}$$

$$\Rightarrow \quad E_t (\hat{c}_{t+1} - \hat{c}_t) = \frac{\tilde{\delta}}{\delta} \underbrace{(E_t \hat{z}_{t+1} - (1-\alpha)\hat{k}_t)}_{\substack{\text{log deviation of} \\ \text{next period's MPK}}}$$

Expected growth in consumption is proportional to expected returns.

$$E_t (\hat{c}_{t+1} - \hat{c}_t) \propto E_t \widehat{MPK}_{t+1}$$

$MPK \uparrow \Rightarrow$ want more consumption tomorrow $\Rightarrow$ save/invest more today $\Rightarrow C$ grows

$\delta \uparrow \Rightarrow$ stronger smoothing, $IES \downarrow$ More curvature $\Rightarrow$ households hate changing C across time

$\delta e \uparrow \Rightarrow$ returns move more with $MPK \uparrow \Rightarrow C$ responds by more

(iii) Technology process: $\log z_{t+1} = (1-\psi) \log \bar{z} + \psi \log z_t + \varepsilon_t$

$$\log(\bar{z} e^{\hat{z}_{t+1}}) = (1-\psi) \log \bar{z} + \psi \log(\bar{z} e^{\hat{z}_t}) + \varepsilon_t$$

$$\Rightarrow \quad \hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t \quad \dots \text{the deviation of technology from S.S. is an AR(1) process}$$

The log-linearized system of equation:

$$\hat{k}_t = \left(\alpha \frac{\overline{Y}}{\overline{R}} + 1 - \delta\right) \hat{k}_{t+1} + \frac{\overline{Y}}{\overline{K}} \hat{z}_t - \frac{\overline{C}}{\overline{K}} \hat{c}_t$$

$$\alpha \frac{\overline{Y}}{\overline{R}} + 1 - \delta = \overline{MPK} + 1 - \delta = \overline{R} = \frac{1}{\beta}$$

$$\frac{\overline{Y}}{\overline{R}} = \frac{\overline{MPK}}{\alpha} = \frac{\overline{R} - (1-\delta)}{\alpha} = \frac{\frac{1}{\beta} - (1-\delta)}{\alpha} = \frac{1 - (1-\delta)\beta}{\alpha\beta} = \frac{\tilde{\delta}}{\alpha\beta}$$

$$\frac{\overline{C}}{\overline{K}} = \frac{\bar{z} \overline{K}^\alpha - \delta \bar{K}}{\bar{K}} = \bar{z} \overline{K}^{\alpha-1} - \delta = \frac{\overline{MPK}}{\alpha} - \delta = \frac{\tilde{\delta}}{\alpha\beta} - \delta$$

$$\Rightarrow \quad \hat{k}_t = \frac{1}{\beta} \hat{k}_{t+1} + \frac{\tilde{\delta}}{\alpha\beta} \hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) \hat{c}_t$$

$$\delta E_t \hat{c}_{t+1} + \tilde{\delta} (1-\alpha) \hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \delta \hat{c}_t$$

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$$

★ Three equations, three variables:
($\hat{k}_t, \hat{c}_{t+1}, \hat{z}_{t+1}$)

Solve the system in terms of exogenous parameters & pre-determined variables. (21)

Pre-determined: $(\hat{z}_t, \hat{k}_{t+1})$

Write in matrix form:

$$\begin{pmatrix} 1 & 0 & 0 \\ \hat{s}(1-\alpha) & \bar{b} & -\hat{s} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{k}_t \\ E_t \hat{c}_{t+1} \\ E_t \hat{z}_{t+1} \end{pmatrix} = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\hat{s}}{\alpha\beta} & \frac{\hat{s}}{\alpha\beta} \\ 0 & \bar{b} & 0 \\ 0 & 0 & \psi \end{pmatrix} \begin{pmatrix} \hat{k}_{t+1} \\ \hat{c}_t \\ \hat{z}_t \end{pmatrix}$$

Because $\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$ is a stationary process, the dynamic property of system is pinned down by the deterministic part: $\hat{z}_t$ is either 0 or converges to 0 since $0 < \psi < 1$

$$M E_t[X_t] = N X_{t-1}$$

where $X_{t-1} = [\hat{k}_{t-1}, \hat{c}_t]'$

$$M = \begin{pmatrix} 1 & 0 \\ \hat{s}(1-\alpha) & \bar{b} \end{pmatrix} \quad N = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\hat{s}}{\alpha\beta} \\ 0 & \bar{b} \end{pmatrix}$$

$$\det(M) = \bar{b} > 0 \Rightarrow M^{-1} \text{ exists} \quad \text{assume } M^{-1} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

$$\text{Impose } MM^{-1} = I \Rightarrow \begin{pmatrix} 1 & 0 \\ \hat{s}(1-\alpha) & \bar{b} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} b_{11} & 0 \\ \hat{s}(1-\alpha)b_{11} + \bar{b}b_{21} & \bar{b}b_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{aligned} b_{11} &= 1 & b_{12} &= 0 \\ b_{21} &= -\frac{\hat{s}(1-\alpha)}{\bar{b}} & b_{22} &= \frac{1}{\bar{b}} \end{aligned}$$

$$\Rightarrow M^{-1} = \begin{pmatrix} 1 & 0 \\ -\frac{\hat{s}(1-\alpha)}{\bar{b}} & \frac{1}{\bar{b}} \end{pmatrix}$$

$$\begin{aligned} \Rightarrow M^{-1}N &= \begin{pmatrix} 1 & 0 \\ -\frac{\hat{s}(1-\alpha)}{\bar{b}} & \frac{1}{\bar{b}} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\hat{s}}{\alpha\beta} \\ 0 & \bar{b} \end{pmatrix} = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\hat{s}}{\alpha\beta} \\ -\frac{\hat{s}(1-\alpha)}{\bar{b}\beta} & 1 + \frac{\hat{s}(1-\alpha)}{\bar{b}}(\frac{\hat{s}}{\alpha\beta} - \delta) \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{\beta} & -\frac{\bar{c}}{\bar{K}} \\ -\frac{\hat{s}(1-\alpha)}{\bar{b}\beta} & 1 + \frac{\hat{s}(1-\alpha)}{\bar{b}}\frac{\bar{c}}{\bar{K}} \end{pmatrix} \end{aligned}$$

Jacobian matrix

$$E_t[X_t] = M^{-1}N X_{t-1}$$

Thus, $M^{-1}N$ shows the property of roots.

$$T = 1 + \frac{1}{\beta} + \frac{\delta(1-\alpha)}{\bar{b}} \frac{\bar{c}}{\bar{K}}$$

$$D = \frac{1}{\beta} \left[1 + \frac{\delta(1-\alpha)}{\bar{b}} \frac{\bar{c}}{\bar{K}} \right] - \frac{\delta(1-\alpha)}{\bar{b}\beta} \frac{\bar{c}}{\bar{K}} = \frac{1}{\beta}$$

$$p(-1) = 1 + D + T = 2 + \frac{2}{\beta} + \frac{\delta(1-\alpha)}{\bar{b}} \frac{\bar{c}}{\bar{K}} > 0$$

$$p(1) = 1 + D - T = -\frac{\delta(1-\alpha)}{\bar{b}} \frac{\bar{c}}{\bar{K}} < 0 \quad \Rightarrow \quad \begin{cases} p(-1) > 0 \\ p(1) < 0 \end{cases}$$

$\Rightarrow$ 1. Stable root in $(-1, 1)$ 1 unstable root in $(1, +\infty)$

Since we have one predetermined variable $\hat{k}_t$ and one jump variable $\hat{c}_t$,

$\Rightarrow$ the system is a saddle path equilibrium.

(From Blanchard and Kahn conditions)

* What is Jacobian Matrix?

Partial derivative that tells how a change in today's state affects tomorrow's variables

eg: $x_t = f(x_{t-1}, z_t)$

$$J = \frac{\partial f}{\partial x} \bigg|_{(x, z)}$$